

EDUARDO LARA PEITER

Machine Learning Engineer & Full-Stack Developer

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PROFESSIONAL SUMMARY

Full-Stack Developer and Machine Learning specialist with 12+ years of experience building scalable software solutions and AI-powered systems. Expert in Python development with proven track record deploying production ML models, real-time data pipelines, and distributed microservices architectures. Experienced in AWS cloud infrastructure, Kubernetes orchestration, and event-driven systems. Currently pursuing postgraduate studies in Artificial Intelligence, combining deep technical expertise with business-driven problem-solving.

TECHNICAL SKILLS

Machine Learning & AI:

TensorFlow, PyTorch, Scikit-learn, Neural Networks, NLP, Computer Vision, Model Deployment, Predictive Analytics, OpenAI API Integration, RAG (Retrieval Augmented Generation), Sentence Transformers

Programming Languages:

Python (Expert), Go, PHP, Java, JavaScript, HTML/CSS, Flutter, TypeScript

Frameworks & Tools:

Django, Flask, FastAPI, React, Angular, Node.js, Git/GitHub, Docker, Kubernetes, Helm, API Integration

Databases:

MySQL, PostgreSQL, MongoDB, ClickHouse, Firebase, Redis (SQL, NoSQL, OLAP)

Cloud & DevOps:

AWS (EC2, S3, RDS, Lambda, API Gateway, CloudWatch, Polly, EKS, ECR), Docker, Kubernetes (K8s), Helm Charts, Google Cloud Platform, Azure, CI/CD Pipelines, Container Orchestration, Agile Methodologies

Data Engineering & Streaming:

Apache Kafka, Apache Flink, ClickHouse, Real-time Data Pipelines (100K+ events/sec), Stream Processing, OLAP Analytics, ETL Pipelines

Message Brokers & Event Systems:

RabbitMQ, Apache Kafka, Event-Driven Architecture, Asynchronous Communication Patterns

Observability & Monitoring:

Prometheus, Grafana, Jaeger, OpenTelemetry (traces, metrics, logs), Distributed Tracing, APM

Architectural Patterns:

Microservices Architecture, Event Sourcing, CQRS, Saga Pattern, Domain-Driven Design (DDD)

Data Science:

Statistical Modeling, Feature Engineering, A/B Testing, Data Visualization

PROFESSIONAL EXPERIENCE

Machine Learning Engineer & Full-Stack Developer

Radio Entre Rios LTDA | January 2013 - Present

- Architected and deployed autonomous news generation system using NLP models on AWS infrastructure, processing 500+ daily news articles with 92% accuracy, reducing manual content creation time by 75%
- Deployed on AWS EC2 with RDS PostgreSQL for scalable database management

- Utilized AWS S3 for content storage and CloudWatch for system monitoring
- Developed AI-powered customer service chatbot using natural language processing, handling 1,200+ monthly user interactions with 87% satisfaction rate
 - Implemented using AWS Lambda and API Gateway for serverless architecture
 - Integrated AWS DynamoDB for conversation history tracking
- Built intelligent audio player with ML-based recommendation engine, increasing user engagement by 34% through personalized content delivery
- Implemented text-to-speech (TTS) pipeline for automated news broadcasting, processing 200+ audio files daily with custom voice models
 - Leveraged AWS Polly for high-quality voice synthesis
 - Automated storage and distribution via AWS S3
- Designed and maintained large-scale database integration for property management platform (arieproperties.com), handling 50,000+ records with optimized query performance
 - Deployed on AWS RDS (MySQL) for high availability
 - Used AWS CloudFront CDN for image delivery optimization
- Created responsive web interfaces for desktop and mobile, improving user experience metrics by 45% through data-driven UX optimization
- Established CI/CD pipelines using GitHub Actions and AWS CodePipeline, reducing deployment time from days to hours and increasing code quality through automated testing
 - Containerized applications using Docker and AWS ECR
 - Orchestrated containerized workloads with Kubernetes on AWS EKS for production deployments
 - Implemented auto-scaling, load balancing, and health monitoring using Kubernetes manifests
- Conducted code reviews and performance optimization, identifying bottlenecks that improved application response time by 38%
- Collaborated with cross-functional teams to translate business requirements into scalable ML solutions, delivering 15+ production-ready applications

Key Technologies: Python, TensorFlow, Scikit-learn, Flask, Django, React, PHP, MySQL, PostgreSQL, AWS (EC2, S3, RDS, Lambda, API Gateway, Polly, CloudWatch, CodePipeline, ECR, EKS), Docker, Kubernetes, Git

EDUCATION

Postgraduate in Artificial Intelligence (In Progress)

Expected Completion: 2026

Postgraduate in Mobile Device Programming

Completed

Bachelor's Degree in Computer Science

Unochapeco, Chapeco, SC | July 1999 - December 2008

KEY PROJECTS

Real-Time Analytics Pipeline

- Built enterprise-grade streaming analytics platform processing 100K+ events/second with sub-second latency
- Implemented Apache Kafka for event ingestion, Apache Flink for stream processing, and ClickHouse for OLAP analytics
- Developed React dashboard with real-time WebSocket updates for live monitoring
- Deployed containerized architecture with Docker Compose for production scalability
- **Tech Stack:** Apache Kafka, Apache Flink, ClickHouse, FastAPI, React, Redis, Docker, WebSocket

E-Commerce Microservices Platform

- Architected distributed microservices system implementing Event-Driven Architecture with Saga pattern for distributed transactions
- Deployed on Kubernetes with Helm charts, featuring auto-scaling, health monitoring, and distributed tracing
- Implemented Event Sourcing and CQRS patterns for data consistency and audit trails
- Integrated Prometheus and Grafana for metrics collection, Jaeger for distributed tracing
- Built services in Python (FastAPI) and Go for high-performance order processing

Tech Stack: Python, FastAPI, Go, Kubernetes, Helm, RabbitMQ, PostgreSQL, MongoDB, Redis, Prometheus, Grafana, Jaeger

SaaS RAG + Observability Platform

- Developed multi-tenant support assistant with Retrieval Augmented Generation (RAG) using OpenAI GPT-4o-mini
- Implemented OpenTelemetry for distributed tracing, Prometheus metrics, and structured logging
- Built tenant isolation with complete data separation and API key authentication
- Created Next.js frontend with FastAPI backend, featuring deterministic fallback when LLM unavailable
- **Tech Stack:** FastAPI, Python, OpenTelemetry, OpenAI API, Sentence Transformers, Next.js, React, TypeScript, Docker

Autonomous News Center Platform

- Developed end-to-end ML pipeline for news aggregation, processing, and generation using Python and NLP models
- Deployed on AWS cloud infrastructure with automated scaling and monitoring
- Integrated web scraping, sentiment analysis, and automated content publishing with 95% uptime
- **Tech Stack:** Python, TensorFlow, BeautifulSoup, Flask, PostgreSQL, AWS (EC2, RDS, S3, CloudWatch)

Property Database Management System

- Built scalable database architecture handling 50K+ property records with complex relational queries
- Implemented RESTful APIs for real-time data synchronization across multiple platforms
- Optimized performance using AWS RDS and CloudFront CDN
- **Tech Stack:** PHP, MySQL, React, REST APIs, AWS (RDS, S3, CloudFront)

Intelligent Audio Recommendation Engine

- Created collaborative filtering algorithm for personalized audio content recommendations
- Achieved 34% increase in user engagement through ML-driven content discovery
- **Tech Stack:** Python, Scikit-learn, Pandas, NumPy

AWS Serverless AI Chatbot (Docker + Kubernetes)

- Built serverless chatbot architecture using AWS Lambda and API Gateway
- Containerized Lambda function with Docker for consistent deployments across environments
- Created production-ready Kubernetes manifests with auto-scaling (HPA), load balancing, and monitoring
- Deployed to AWS EKS with ECR for container registry
- Implemented natural language processing for intelligent conversation

handling • Integrated conversation persistence with AWS DynamoDB • **Tech Stack:** Python, NLP, AWS (Lambda, API Gateway, DynamoDB, EKS, ECR), Docker, Kubernetes

Automated Text-to-Speech Broadcasting System

• Developed pipeline for automated news audio generation using AWS Polly • Processes 200+ daily audio files with automated storage and distribution • **Tech Stack:** Python, AWS (Polly, S3, Lambda)

CORE COMPETENCIES

- Machine Learning Model Development & Deployment
- Real-Time Data Pipeline Engineering (Kafka, Flink)
- Microservices Architecture & Event-Driven Systems
- Full-Stack Web & Mobile Application Development
- Natural Language Processing & Text Analytics
- Database Design & Optimization (SQL, NoSQL, OLAP)
- Cloud Infrastructure & AWS Services
- Container Orchestration with Kubernetes & Helm
- Observability & Distributed Tracing (OpenTelemetry, Jaeger)
- Docker Containerization & Multi-stage Builds
- Agile Project Management & Team Collaboration
- API Integration & RESTful Services
- DevOps & CI/CD Implementation
- Code Versioning & Quality Assurance

LANGUAGES

Portuguese: Native

English: Proficient (C2 Level)